ANSWER KEY & MARKING SCHEME SET 1

Chapter 12: **Reproduction: How Life Continues** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(c)** 14 March	Mid cycle.
3	**(b)** the one with an eye grows	Bud needed.
4	**(c)** wings or hairs	Light structures.
5	**(b)** oviduct	Tube.
6	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
7	**(c)** 13 to 15 March	Around ovulation.
8	**(c)** cambia of stock and scion are in contact	Tissue contact.
9	**(c)** hooks or fleshy fruit	Burrs.
10	**(b)** prostate and seminal vesicles	Semen.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Muscle 46, sperm 23 (1); Egg 23, zygote 46 (1).	1+1
12	Pollen grains carry male gametes (1); Ovule has the egg; becomes seed after fertilisation (1).	1+1
13	IVF (test-tube baby) (1); Artificial insemination (1).	1+1
14	Buds (eyes) grow using stored food (1); Vegetative reproduction of a tuber (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Ovulation about 14 days before the next period: day 16 (1); Fertile period: days 13 to 17 (1); Sperms survive 3 days before (1).	1+1+1
16	Seeds from sexual reproduction give variation (1); Cutting is a clone (1); Mitosis vs meiosis and fertilisation (1).	1+1+1
17	Unisexual flowers (1); Anthers and stigma mature at different times (1); Different positions (1).	1+1+1
	OR: Frog: thousands of eggs, no care (1); Hen: few eggs, some care (1); Human: one baby, extensive care (1).	1+1+1
18	Retains the quality of fruit (1); Faster bearing of fruit (1); Combines good stock and scion traits (1).	1+1+1
	OR: Poor nutrition and education (1); Pressure on resources (1); Unemployment (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Tube 1: moist cotton, room temperature (1); Tube 2: dry cotton (1); Tube 3: seeds submerged in boiled water with oil (1); Tube 4: moist cotton in a fridge (1); Only tube 1 germinates; conclusion (1).	1+1+1+1+1
	OR: Causes: blocked tubes, low sperm count (1); Hormonal problems (1); IVF steps (1); Artificial insemination (1); Social acceptance and ethics (1).	1+1+1+1+1
20	Testes, ovaries (1); Oviduct and fertilisation (1); Implantation (1); Placenta and cord (1); Birth (1).	1+1+1+1+1
	OR: Nutrition (1); STIs and prevention (1); Substance abuse effects (1); Hygiene (1); Awareness and medical care (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 12: **Reproduction: How Life Continues** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(a)** chance of fertilisation	Fertility.
3	**(d)** twig of the desired plant	Grafted part.
4	**(b)** water	Floating husk.
5	**(d)** epididymis	Mature.
6	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
7	**(a)** 1/2	Independent events.
8	**(d)** the root system	Base.
9	**(b)** maize	One cotyledon.
10	**(d)** sperm production needs a lower temperature	Slightly lower.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Different eggs and sperms (1); Different genes (1).	1+1
12	Individual can live without reproducing (1); Species ends if none reproduce (1).	1+1
13	Brings variation (1); Helps in adaptation (1).	1+1
14	Lack of oxygen (1); Seed needs air for respiration (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Seeds = 50 (1); Seedlings = 20 (1); 40 per cent of 50 (1).	1+1+1
16	Lining retained by progesterone (1); Uterus enlarges (1); Oxytocin causes contractions (1).	1+1+1
17	Sex-selective abortion is illegal (1); Equality of girls and boys (1); Counselling and law (PCPNDT) (1).	1+1+1
	OR: Coconut: water (1); Maple and cotton: wind (1); Xanthium: animals; balsam: explosive (1).	1+1+1
18	Bisexual: both stamen and carpel, e.g. hibiscus, mustard (1); Unisexual: only one, e.g. papaya, cucumber (1); Pollination differences (1).	1+1+1
	OR: Hen: internal; sperm enters the female (1); Frog: external; sperm released in water (1); Number of eggs differ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Fission and budding (1); Fragmentation and regeneration (1); Spores (1); Vegetative propagation (1); Advantages: fast; disadvantages: no variation (1).	1+1+1+1+1
	OR: Mother XX, father XY (1); Gametes: X; X or Y (1); Combinations XX, XY (1); Probability 50 per cent each (1); Sperm decides (1).	1+1+1+1+1
20	Seeds are from sexual reproduction with variation (1); Genes recombine (1); Use grafting or budding (1); Steps of grafting (1); Clones keep traits (1).	1+1+1+1+1
	OR: Plant: pollen and ovule, pollination, fertilisation inside the ovule (1); Frog: sperm and egg meet in water (1); Plant: seed and seedling; frog: tadpole (1); Plants disperse seeds, frogs move as adults (1); Both show alternation of generations of stages (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 12: **Reproduction: How Life Continues** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(a)** 1 : 1	Equal X and Y sperms.
3	**(a)** branch while it is still attached to the parent	Rooting on the stem.
4	**(a)** low water content and food stored	Dormancy.
5	**(b)** progesterone	After ovulation.
6	**(c)** A is true but R is false.	A is true; R is false because each spore has a low chance.
7	**(b)** girl	XX.
8	**(b)** banana	Seedless.
9	**(c)** cotyledon	Two cotyledons.
10	**(d)** ovary	Ovulation.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Sperms swim to the egg (1); No pollen tubes (1).	1+1
12	Sexual: passed on with variation (1); Asexual: the clone carries the same (1).	1+1
13	No hCG from embryo (1); Progesterone falls and menstruation occurs (1).	1+1
14	Dispersed by air (1); Reach new food sources (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Mitosis: body cells, 2 cells (1); Meiosis: gonads, 4 cells (1); Chromosome number maintained vs halved (1).	1+1+1
16	Planaria: whole body from a piece (1); Humans: limited, tissue repair (1); Cells divide and differentiate (1).	1+1+1
17	Dicot seed (1); Radicle to root; plumule to shoot (1); Cotyledons give food; seed coat protects (1).	1+1+1
	OR: Sperm: tail for swimming, mitochondria (1); WBC: engulf germs (1); Nerve: carries impulses (1).	1+1+1
18	Gametes have half the number (1); Fertilisation restores it (1); Meiosis and fertilisation (1).	1+1+1
	OR: Protects seeds and helps dispersal (1); Food for humans (1); Source of seeds and economic crops (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Pollen germinates on the stigma (1); Pollen tube grows to the ovule (1); Double fertilisation (1); Ovule becomes seed (1); Ovary becomes fruit (1).	1+1+1+1+1
	OR: Gametes and variation (1); Asexual: clones (1); Sexual: mixing of genes (1); Natural selection acts on variation (1); Evolution of species (1).	1+1+1+1+1
20	Days 1 to 5: lining is shed (menstruation) (1); About day 14: an egg is released from the ovary (1); Lining thickens to receive the embryo (1); Fertilisation: embryo implants, no menstruation (1); No fertilisation: lining is shed and the cycle repeats (1).	1+1+1+1+1
	OR: Select flowers and a time (1); Count visitors (1); Record in a table (1); Analyse the most common pollinators (1); Conservation: gardens, reduce pesticides (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.